

Data-Quality Brief: CAA Air Datasets

Every figure below is recomputed from the raw data by `scripts/data_quality/data_quality_audit.R` (summary written to `output/explore_tabulations/data_quality/data_quality_summary.csv`). “Bad dates” are values that parse to a calendar year outside the plausible window **1965–2026**; “un-parseable” are non-blank date values that do not parse to a date at all. Missingness is the share of rows that are blank or NA.

Summary

Table	Rows	Worst missingness	Bad dates	Other flags
ICIS-Air				
FACILITIES	279,211	LOCAL_CONTROL_REGION 95.8%	0	key unique
PROGRAMS	456,601	—	36	
PROGRAM_SUBPARTS	190,570	—	—	
POLLUTANTS	976,479	CAS number 60.1%	—	
FCES_PCES	1,802,044	PROGRAM_CODES 41.5%	23	ACTIVITY_ID not row-unique (626)
VIOLATION_HISTORY	101,147	AIR_LCON_CODE 81.3%	5	key unique
FORMAL_ACTIONS	105,656	—	0	ACTIVITY_ID dup 2,911; penalty skew
INFORMAL_ACTIONS	336,410	—	133	ACTIVITY_ID not row-unique (164,256)
STACK_TESTS	646,332	POLLUTANT_DESCS 100%	8	key unique
TITLEV_CERTS	2,563,435	deviation flag 56.9%	60	ACTIVITY_ID not row-unique (2,076,220)
CAA Pipeline				
PIPELINE	66,655	FEA_ISSUE_DATE_FLAG 99.7%	1	42,552 un-parseable end-dates; 10.8% placeholder rows; comp-cost 99.4% zero

AFS (legacy, frozen 2014)

Table	Rows	Worst missingness	Bad dates	Other flags
AFS_FACILITIES	236,734	CURRENT_HP_V 99.4%	—	key unique
AIR_PROGRAM	1,139,429	CAS number 94.2%	—	undocumented codes G, R (623 rows)
AFS_ACTIONS	2,579,661	viol. type codes 99.5%	30	penalty 98.2% zero
AFS_AIR_PRG_HIST_COMPLIANCE	10,204,801	— (complete)	—	
AFS_HP_V_HISTORY	32,057	HP_V_RESOLVED_* 6.6%	—	6.6% episodes unresolved
Emissions				
POLL_RPT_COMBINED_EMISSIONS	19,411,871	NEI_HAP_VOC_FLAG 44.0%	—	emission max 47.6 billion; 5.1% zero
FRS crosswalk				
AFS → FRS	236,734	—	—	71,799 (30.3%) AFS facilities unmatched (state-driven)

ICIS-Air

- **Local fields are essentially empty.** LOCAL_CONTROL_REGION_CODE/NAME are 95.8% missing in FACILITIES; AIR_LCON_CODE is 81.3% missing in VIOLATION_HISTORY.
- **ACTIVITY_ID is not a row-level key in several tables.** It repeats across rows in TITLEV_CERTS (only 487,215 distinct of 2,563,435 rows), INFORMAL_ACTIONS (172,154 of 336,410), FORMAL_ACTIONS (2,911 duplicate rows) and FCES_PCES (626) — one activity spans multiple program/pollutant rows. De-duplicate on ACTIVITY_ID before counting events.
- **STACK_TESTS POLLUTANT_DESCS is 100% missing** (and POLLUTANT_CODES 37.0%); pollutant detail is effectively unavailable for stack tests.
- **Penalty distribution is extreme and right-skewed.** FORMAL_ACTIONS.PENALTY_AMOUNT ranges to \$100,728,768 with 31.6% of actions at zero.
- **Sparse date coverage in places** — e.g. DSCV_PATHWAY_DATE 62.0% and HPV_DAYZERO_DATE 56.0% missing in violations. Small numbers of out-of-range dates appear in most event tables (FCES 23, INFORMAL 133, TITLEV 60).

CAA Pipeline

- **Placeholder rows.** 7,193 rows (10.8%) carry system-generated VIOL_ACTIVITY_IDs (9906/9913 prefixes) rather than real violations.
- **VIOL_END_DATE is a mixed text field.** 42,552 values are non-dates such as “Unresolved” / “N/A”; only one parses to an out-of-range year.
- **Compliance-action cost is near-empty.** EA_COMP_ACTION_COST is zero on 99.4% of rows (non-zero on 94); EA_PENALTY_AMT is missing on 74.6% and zero on 14.8% (max \$47.8M).

- FEA_ISSUE_DATE_FLAG is 99.7% missing.

AFS (legacy, frozen 2014)

- **Undocumented program codes.** AIR_PROGRAM_CODE contains the values G and R (623 rows) that are not in EPA’s documented program-code list.
- **Very high missingness in detail fields.** CURRENT_HP_V 99.4% and SECONDARY_SIC_CODE 93.4% missing in FACILITIES; ALL_VIOLATION_TYPE_CODES 99.5% and ALL_VIOLATING_POLL_CODES 94.8% in ACTIONS; CAS number 94.2% in AIR_PROGRAM.
- **Penalties are rare.** AFS_ACTIONS.PENALTY_AMOUNT is zero on 98.2% of actions (max \$24M); a small number of DATE_ACHIEVED values are out of range (28) or unparseable (2).
- **The quarterly compliance table is complete** (AFS_AIR_PRG_HIST_COMPLIANCE, 10.2M rows, no missing status); HPV episodes are 6.6% unresolved.

Emissions

- ANNUAL_EMISSION is fully populated (0% missing) but spans an enormous range (max 47,629,708,350 in mixed units — pounds vs. MTCO_{2e}), with 5.1% reported as zero and no negative values. Mixed units mean values cannot be summed across programs without conversion.
- NEI_HAP_VOC_FLAG is 44.0% missing and NEI_TYPE 9.9%.

FRS crosswalk

- 71,799 AFS facilities (30.3%) do not link to an FRS REGISTRY_ID via the identifier crosswalk. The gap is strongly state-specific (near-zero match in some states, near-complete in others) rather than driven by facility closure; most unmatched facilities still appear in FRS/ICIS by name and address. See the AFS–ICIS–Air Crosswalk Brief for detail.

Sources

- All figures: scripts/data_quality/data_quality_audit.R (and output/explore_tabulations/data_qualit
- Cross-referenced against the per-table exploration notes and the Pipeline / AFS–ICIS–Air Crosswalk briefs.